

Fossil Fuel Prices and Capital Cost: A Machine Learning Driven Study on Energy Transition



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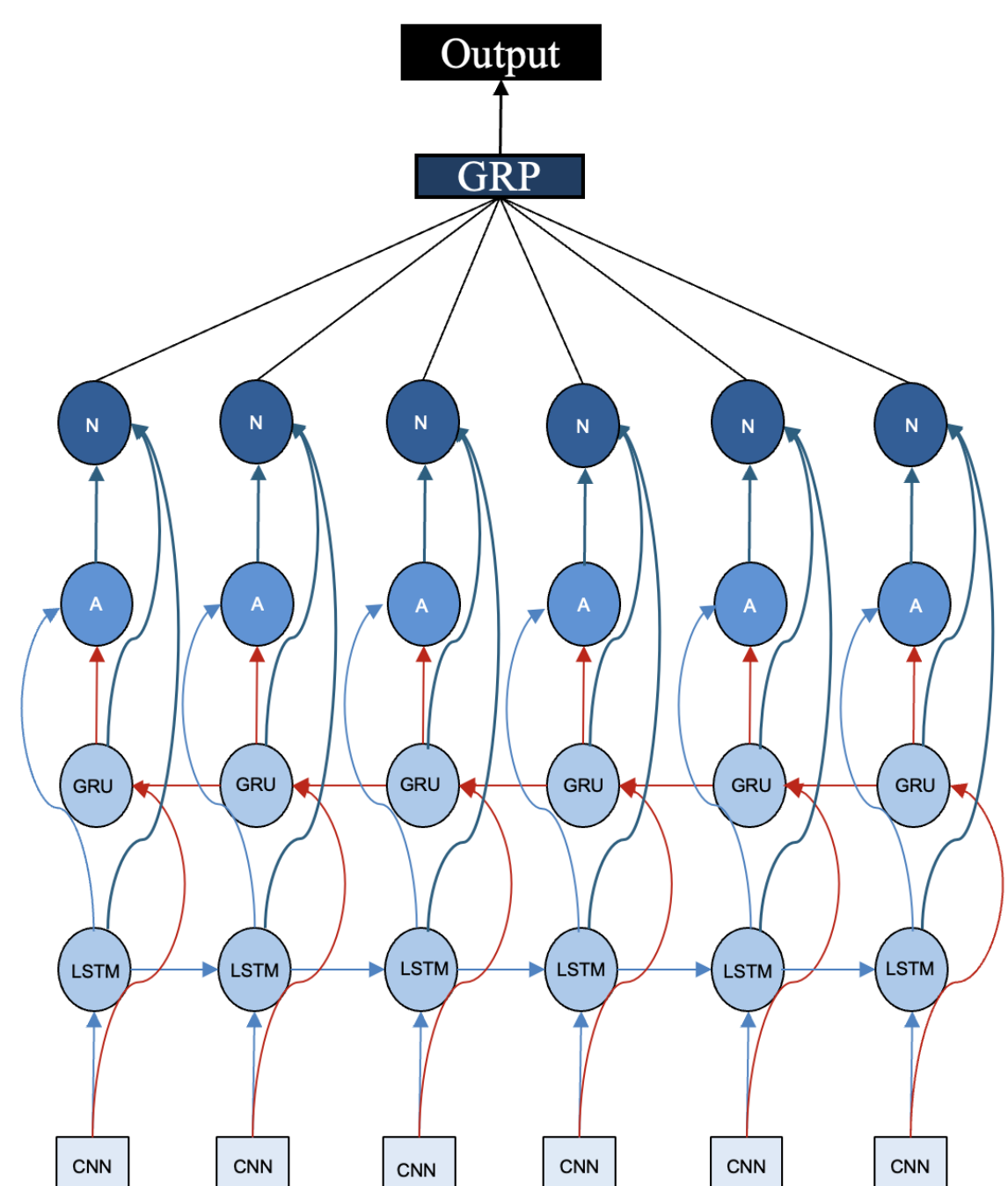
Introduction

Achieving the 1.5°C climate target necessitates a rapid energy transition, yet this process faces financial barriers. Renewable projects remain vulnerable to shocks from the legacy fossil fuel system. Therefore, this study focuses on the upcoming decade and seeks to answer the following questions:

- RQ1:** How do fossil fuel price fluctuations affect the WACC of energy transition projects, influencing the pace of renewable energy deployment?
- RQ2:** How do these price dynamics affect capital costs across regions, particularly between fuel-importing and fuel-exporting countries, and how should financing strategies be shaped?

Methodology

Stage 1: Fossil Fuel Price Prediction We employ a hybrid model combining Long Short-Term Memory (LSTM) for capturing temporal dependencies and Gaussian Regression Process (GRP) for probabilistic refinement.



The final price P_e is refined by the GRP kernel to smooth the LSTM predictions (y_{DL}):

$$P_e = K(t, X)[K(X, X) + \sigma_n^2 I]^{-1} y_{DL} \quad (1)$$

Stage 2: Inflation Rate Calculation We model it as a non-linear function using an ensemble of LSTM networks feeding into a Feed-Forward network:

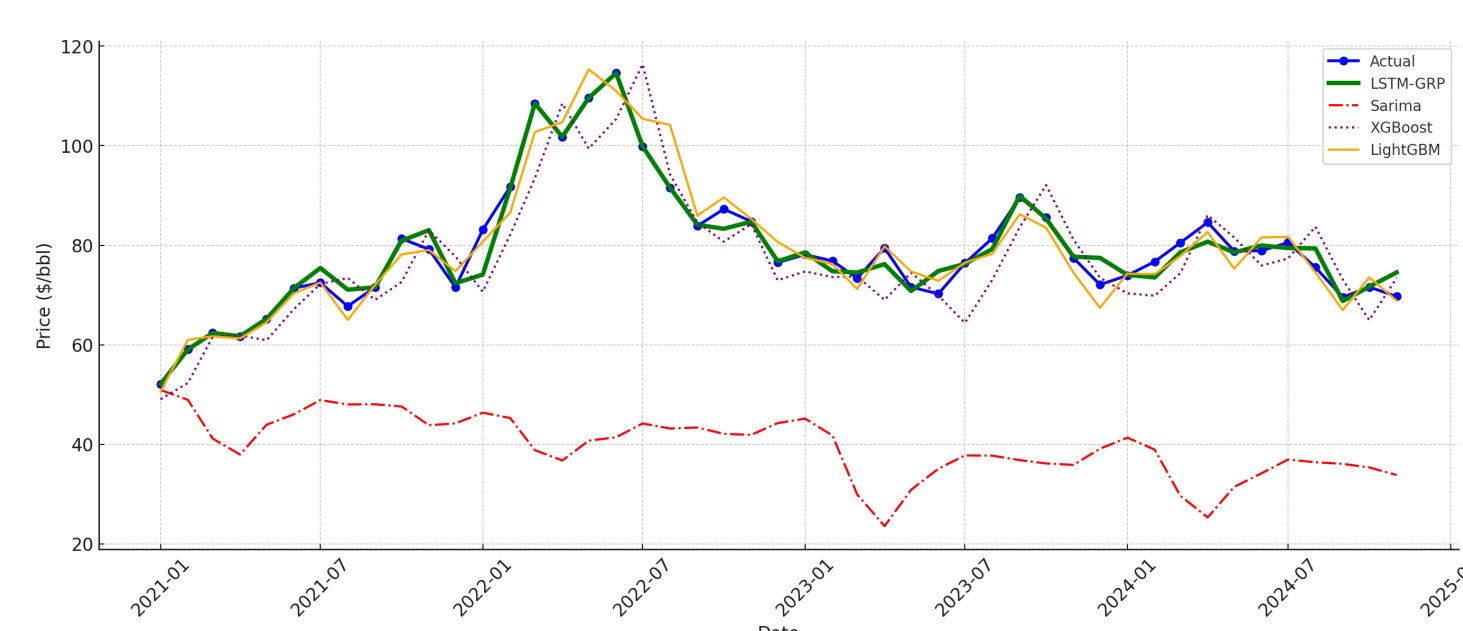
$$\hat{y}_{t+h} = g_{FF}(LSTM(x_t; \Theta_{LSTM}); \Theta_{FF}) \quad (2)$$

Stage 3: WACC Impact Evaluation The model involves three fossil fuel prices (Coal, Natural Gas, and Oil) and nine types of energy transition projects. We train 27 independent regression models across 9 energy sectors. The model seeks to find function f quantifying the specific impact of price (P_e) and inflation ($\hat{y}_{ens}$):

$$WACC_p = f(P_e, \hat{y}_{ens, t}) + \epsilon \quad (3)$$

The approach utilizes *Auto Stacking* to combine base models for optimal predictive performance (R^2 , MAPE).

Model Performance Evaluation



Accuracy of Prediction Model

This study employed four models to forecast WTI crude oil prices from 2021 to 2024. Evaluation metrics revealed that the LSTM-GRP model achieves a coefficient of determination of 0.966. Additionally, it exhibited lower RMSE and MAE values of 2.105 and 1.6342. Furthermore, the accuracy of forecasting the inflation rate has been examined in previous studies.

Main Results

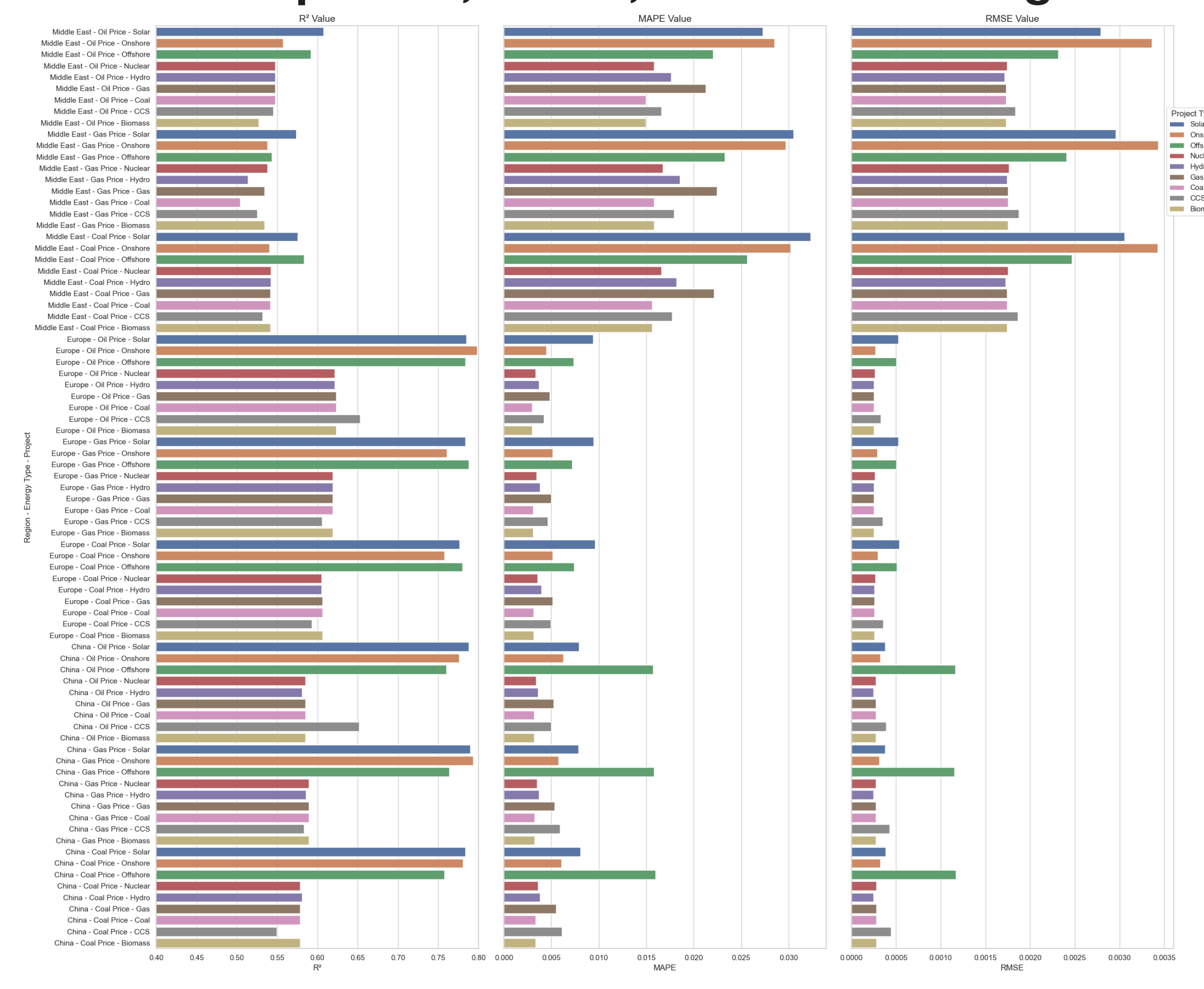
Sectoral Variances

Solar and Wind projects are most vulnerable to price shocks, with average $R^2 > 0.7$. While traditional sectors (Hydro, Biomass, Coal) show weaker correlations.

Regional Heterogeneity

- Importing Regions (Europe & China):** Strong influence ($R^2 \in [0.65, 0.67]$) and low error (MAPE < 0.006) indicate direct risk transmission from global markets.
- Exporting Region (Middle East):** Weaker influence ($R^2 = 0.5$) but higher uncertainty (MAPE=0.021), reflecting the decoupling effect of fiscal buffers..

Heatmap of R2, MAPE, RMSE across region



Analysis

Fossil Fuel Prices $\rightarrow$ WACC

WACC is the weighted average cost of debt:

$$WACC_{it} = L_{it} r_{Dit} (1 - Tax_{it}) + (1 - L_{it}) r_{Eit}$$

Fossil fuel prices could impact WACC through the following two forces:

- Cost of Equity (r_E)**
 - The Substitution Effect:** High fossil fuel prices render renewable energy more cost-competitive, reducing the required return on equity (r_E).
 - Systematic Risk (β):** Persistent volatility increases market uncertainty, raising the risk premium for the entire energy sector.
- Cost of Debt (r_D)**
 - The Inflation Link:** It drives global inflation, compelling central banks to raise interest rates, increasing borrowing costs for capital-intensive projects.
 - Credit Spreads:** Conversely, high energy prices improve the valuation of fixed-price Power Purchase Agreements, reducing default risk and credit spreads.

Table 1: Theoretical Channels of Price Transmission

Price Signal	Affected Component	Transmission Channel
Level Increase	r_E (Cost of Equity)	Substitution Effect
Level / Volatility	r_E (Cost of Equity)	Market Correlation
Volatility Increase	r_E (Cost of Equity)	Hedging Property
Volatility Increase	r_E (Cost of Equity)	Climate Risk Pricing
Volatility Increase	r_D (Cost of Debt)	Macroeconomic Contagion
Level Increase	r_D (Cost of Debt)	Off-taker Stability
Level Increase	r_D (Cost of Debt)	PPA Contract Value
Volatility Increase	r_D (Cost of Debt)	Debt Market "Green Premium"

Discussion

1. Drivers of Regional Heterogeneity

- Europe:** High import dependence (82%) and geopolitical exposure amplify external shocks ($R^2 = 0.67$).
- China:** The "Price Corridor" mechanism and fiscal subsidies dampen market signals, artificially stabilizing capital costs ($R^2 = 0.64$) despite high import needs.
- Middle East:** Sovereign wealth funds create a "policy isolation layer." This *Fiscal Cushion* ensures low capital costs ($R^2 = 0.54$) but reduces the economic urgency for decarbonization.

2. Policy Implications for 1.5°C Target

Financial risk translates to environmental risk. Tailored strategies are essential:

- For Importers (EU/CN):** Mitigate volatility via **Green Bonds** and climate insurance. China should transition toward market-based pricing to reflect true green premiums.
- For Exporters (ME):** Leverage sovereign capital to proactively diversify energy assets before the fiscal cushion erodes, preventing "transition inertia."

Acknowledgments

The Duke Kunshan University Education Development Foundation supports the research. The research results of this article are also sponsored by the Kunshan Municipal Government research funding.